

MANTRA REGISTERED DEVICE SERVICE API WINDOWS

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1. What is Mantra Registered Device Service?

UIDAI has discontinued the support of public device with latest authentication API 2.0 (rev 1). With this new API, user must have to use registered device to perform successful authentication request. RD service comes with this support so user can perform authentication transaction.

2. About

The document provides the functional and implementation information to work with Aadhaar Registered device MFS100 (Mantra Fingerprint Sensor). By using this service, you can capture fingerprint from MFS100. This SDK provided facility to capture encrypted PID data of user/resident that will be used for authentication using Aadhaar authentication API 2.0 (rev 1).

3. Regulation

RD service functionalities implemented as per the guidelines provided by UIDAI to comply with the requirements. No further customization will be provided with this.

4. Requirement

Supporting OS: Windows 7/8/10

Client Setup: MantraRDSERVICE.exe (Administrator privileges required while installation)

Internet Connectivity with Mantra Server: As per RD service architecture declared by UIDAI, it is mandatory to have connectivity with vendor (mantra) server for device validation and such operations.

5. API Functionalities

Generally, RD service having only three functionalities can be call over VERB * HTTP/1.1 request only such as—

- ✓ Device Discovery
- ✓ Device Information
- ✓ Capture Biometric

5.1 Device Discovery

RD service can be run on any of port from range 11100 to 11120. Client application must have to scan all ports to found installed RD service in the computer.

Following is the request format:

```
RDSERVICE * HTTP/1.1  
HOST: http://127.0.0.1:[port]  
EXT: APP_NAME
```

The response is as follows:

HTTP/1.1 200 OK
CACHE-CONTROL:no-cache
LOCATION:http://127.0.0.1:<rd_service_port>
Content-Length: length in bytes of the body
Content-Type: text/xml
Connection: Closed

```
<RDService status="READY|USED|NOTREADY|..." info="provider info for display purposes">  
<Interface id="CAPTURE" path="/rd/capture" />  
<Interface id="DEVICEINFO" path="/rd/info" />  
</RDService>
```

Based on the path, subsequent calls from application can be made to RD Service.

<http://127.0.0.1:<port>/<CAPTURE-path>>
<http://127.0.0.1:<port>/<INFO-path>>

Note:

RD Service – Http and Https Ports are same on 11100 to 11120.
RD Service - Https Port is on 8005 will be removed from next Version.

5.2 Device Information

Following is the request format:

DEVICEINFO http://127.0.0.1:<rd_service_port>/<INFO_path>
HOST: 127.0.0.1:<port>

The response is as follows:

HTTP/1.1 200 OK
CACHE-CONTROL:no-cache
LOCATION:http://127.0.0.1:<rd_service_port>/<INFO_path>
Content-Length: length in bytes of the body
Content-Type: text/xml
Connection: Closed

```
<DeviceInfo dpld="" rdsId="" rdsVer="" dc="" mi="" mc="" />
```

dpld – (mandatory) Unique code assigned to registered device provider.
rdsId – (mandatory) Unique ID of the certified registered device service.
rdsVer – (mandatory) Registered devices service version.
dc – (mandatory) Unique Registered device code.
mi – (mandatory) Registered device model ID.

mc – (mandatory) This attribute holds registered device public key certificate.
This is signed with device provider key.

5.3 Capture Biometric

The capture call is a blocking call and only one client can call at any point in time.

Following is the request format:

CAPTURE http://127.0.0.1:<rd_service_port>/<CAPTURE_path>
HOST: 127.0.0.1:<port>

```
<PidOptions ver="">
<Opts fCount="" fType="" iCount="" iType="" pCount="" pType="" format="" pidVer="" timeout=""
otp="" wadh="" posh="" env=""/>
<Demo></Demo>
<CustOpts>
<!-- no application should hard code these and should be configured on app or AUA servers. These
parameters can be used for any custom application authentication or for other configuration
parameters. Device providers can differentiate their service in the market by enabling advanced
algorithms that applications can take advantage of. -->
<Param name="" value="" />
</CustOpts>
</PidOptions>
```

PidOptions:

ver: Version of the PidOptions spec. Currently it is "1.0". This is necessary to allow applications to gracefully upgrade even when RD service may be been upgraded.

Opts:

Int fCount (optional) number of finger records to be captured (0 to 10)
Int fType (optional) ISO format (0 for FMR, 1 for FIR, 2 for both), 0 (FMR) is default
Int iCount (optional) number of iris records to be captured (0 to 2)
Int iType (optional) ISO format (0 for IIR), 0 (IIR) is default
Int pCount (optional) number of face photo records to be captured (0 to 1). Currently face matching is not supported.
Int pType (optional) face format. Currently face matching is not supported.
Int format (mandatory) 0 for XML, 1 for Protobuf
String pidVer (mandatory) PID version
Int timeout capture timeout in milliseconds
String otp (optional) OTP value captured from user in case of 2-factor auth
String wadh (optional) **If passed, RD Service should use this within PID block root element "as-is".**
String env (optional) UIDAI Authentication environment for which capture is called. Valid values are "P" (Production), "PP" (Pre-Production), and "S" (Staging). If blank or if the attribute is not passed, RD service should default this to "P". This is provided to allow same RD service to use different UIDAI public key based on the environment.

String posh (optional) if specific positions need to be captured, applications can pass a comma delimited position attributes. See “posh” attribute definition in Authentication Specification for valid values. RD Service (if showing preview) can indicate the finger using this. If passed, this should be passed back within PID block. Default is “UNKNOWN”, meaning “any” finger/iris can be captured.

Demo:

Element allows demographic data to be passed to form PID block as per authentication specification

CustOpts:

mantrakey (optional) unlocked key, if device is locked with specific key

The response is as follows:

HTTP/1.1 200 OK

CACHE-CONTROL:no-cache

LOCATION:http://127.0.0.1 :<rd_service_port>/<CAPTURE_path>

Content-Length: length in bytes of the body

Content-Type: text/xml

Connection: Closed

<PidData>

<Resp errCode="" errInfo="" fCount="" fType="" iCount="" iType="" pCount="" pType="" nmPoints="" qScore="" />

<DeviceInfo dpld="" rdsId="" rdsVer="" dc="" mi="" mc="" />

<Skey ci="">encrypted and encoded session key</Skey>

<Hmac>SHA-256 Hash of Pid block, encrypted and then encoded</Hmac>

<Data type="X|P"> *base-64 encoded encrypted pid block* </pid>

</PidData>

Resp:

Int errCode (mandatory) 0 if no error, else standard error codes

String errInfo (optional) additional info message in case of error/warning

Int fCount (mandatory for FP) number of finger records actually captured

Int fType (mandatory for FP) actual format type – 0 (FMR) , 1 (FIR) , 2(BOTH)

Int iCount (mandatory for Iris) number of iris records actually captured

Int iType (mandatory for Iris) actual Iris format (0 for IIR)

Int pCount (mandatory for Photo) number of face photo records actually captured. Currently face matching is not supported.

Int pType (mandatory for Photo) face format. Currently face matching is not supported.

Int nmPoints (mandatory for FMR capture) Number of minutiae points when FMR is captured. Applications may use this for accepting or retrying the capture. If multiple fingers are captured, send comma delimited numbers.

Int qScore (optional) If quality check is done, send a normalized score that is between 0 and 100.

Skey:

String skey (mandatory) encrypted session key as per auth spec

String ci (mandatory) UIDAI public key identifier as per auth spec

Hmac:

String hmac (mandatory) hmac value as per auth spec

DeviceInfo:

Device info as per discuss earlier in this document.

6. REFERENCE

For more information, please refer below documents:

- aadhaar_registered_devices_2_0_2.pdf
- aadhaar_authentication_api_2_0_1.pdf

7. DEVELOPMENT SUPPORT

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malay.pandya@mantratec.com

Please note, this contact information is only for development supports and it should not be shared to your clients or end user.

8. TECHNICAL SUPPORT

Mantra Support Team

+91-79-49068000

servico@mantratec.com

This information can be shared with your clients or end user for any kind of technical support.

9. ENABLE DEVELOPMENT DEVICES

To enable development device over staging or pre-production, send serial number of device to

servico@mantratec.com

Please mentioned that in which environment (staging or pre-production) you need to add the devices.

Note: At a time of production, all devices will automatic enabled.

-X-X-X-X-X-X-X-X-X-X-X-X-X-X-X-X-X-